

# Anthony Retelewski

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## EDUCATION

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Case Western Reserve University

Expected: May 2028

*BS, Computer Science, Minor in Math and Mechanical Design*

*Cleveland, Ohio*

- **Relevant Coursework:** Data Structures & Algorithms, Logic Design & Computer Organization, Linear Algebra, Full-Stack Web Development

## EXPERIENCE

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### CWRU Quants

March 2025 - Present

*Quantitative Research Lead*

*Cleveland, Ohio*

- Engineered a sentiment analysis data ingestion pipeline using Postgres for 200,000+ earnings call documents
- Developed and deployed Natural Language Processing (NLP) models to perform sentiment analysis and identify key market-relevant topics
- Architected a sophisticated feature engineering process using techniques like TF-IDF and word embeddings to convert linguistic patterns into quantitative features for machine learning models
- Developed and evaluated dozens of quantitative metrics to capture linguistic and market-relevant patterns
- Applied machine learning models to identify predictive signals from transcript-derived features

### CWRU Motorsports

Aug. 2024 - Present

*Design Engineer*

*Cleveland, Ohio*

- Led complete redesign of the team website using Vite and TypeScript, increasing site traffic by 18% and improving performance, maintainability, and user experience
- Co-developed a TypeScript-based inventory tracking system to streamline team stock organization
- Designed, manufactured, and validated rear drivetrain yokes
- Supported brakes subteam in the design, manufacturing, and testing of brake rotors and pistons
- Developed a Discord bot to summarize messages over specific timeframes, helping users catch up quickly

### CWRU Club Tennis

Aug. 2025 - Present

*Tournament Coordinator*

*Cleveland, Ohio*

- Organized tournament logistics, including scheduling, travel, and rosters
- Designed and launched custom club merchandise, incorporating team branding and member feedback

### FTC Robotics - Team 15118

Aug. 2022 - March 2024

*Team Captain, Lead Programmer*

*Deerfield, Illinois*

- Designed, manufactured, and tested custom robotics components using CAD software and 3D printing
- Programmed and tuned complex autonomous pathing/localization system in Java, printed and built custom odometry pod system
- 2024 state semifinalists, 2x Design Award 1st place, Inspire Award 3rd Place

## PROJECTS

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### Stock Price Estimation

*Columbia University Summer Program*

- Developed a deep learning model in Python to estimate future stock price movements from time-series data
- Integrated sentiment analysis to improve model accuracy and confidence
- Utilized hundreds of live financial data points from Yahoo Finance API for training and validation

### Custom FPV Drones

- Extensive experience in custom building, flying, and programming FPV drones
- Programmed microelectronics using Betaflight firmware
- Developed autonomous return-to-home flight capabilities

## CERTIFICATIONS, SKILLS & INTERESTS

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- Certifications: SolidWorks CAD Design Associate
- Technologies: Python, Java, Git, JavaScript, TypeScript, SQL Pandas, Numpy, Siemens NX, SolidWorks
- Skills: Critical-thinking, communication, collaboration, adaptability, attention to detail
- Languages: English (native), Polish (fluent)
- Interests: Tennis, Climbing, Drones, Photography, Wakeboarding